All fixed joint words above a Slater state

Local CAR polynomials, the same-disorder limit, and finite-time evolution

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Abstract

We extend the same-disorder Slater calculation from fourth moments to every fixed joint word of the Hamiltonian and its selected particle–hole block. An exact single-site polynomial retains CAR signs, including negative finite-size six-point contributions. A coefficient estimate controls all multiple-overlap configurations after normalization. Only quadratic site clusters survive, while backward selected edges vanish. The resulting positive word law constructs the joint cyclic representation. Uniform finite-size row/column covariance bounds then control the complete time series, giving annealed finite-time correlators and a self-adjoint limiting Hamiltonian. This is a model-level joint dynamics result; the stationary physical state and the de Sitter bulk dictionary remain separate targets.

1 The model and what is being extended

Use N complex fermions with standard CAR, $m = N/2$, $a = N - m = m$, and a fixed occupied set O, empty complement E, and $\Omega_O = |O\rangle$. With increasing-order creators and adjoint annihilators, set

$$H_N = \sqrt{w_N} \sum_{|A|=|B|=p} J_{AB} C_A^\dagger C_B, \quad w_N = \frac{4^p J_s^2}{\lambda_N} \left(\frac{p!}{N^p} \right)^2, \quad \lambda_N = \frac{p^2}{N}. \tag{1}$$

The centered Hermitian Gaussian matrix obeys $\mathbb{E} J_{AB} J_{CD} = \delta_{AD} \delta_{BC}$; overlapping A, B are included. Write $\varphi_N(X) = \mathbb{E}\langle \Omega_O, X \Omega_O \rangle$. All operators below use this same J. Take $p \to \infty$, $\lambda_N \to \lambda > 0$, fixed $J_s > 0$, $\hbar = 1$ and string units for H_N. The microscopic state is not assumed stationary.

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The normalized operators and exact counts are

$$L_N = \binom{m}{p}\binom{a}{p}, \qquad D_N = \binom{m}{p}\binom{a+p}{p}, \tag{2}$$

$$h_N = H_N/\sqrt{w_N D_N}, \qquad M_N^\dagger = L_N^{-1/2} \sum_{P\subset E,\ Q\subset O} J_{PQ} C_P^\dagger C_Q. \tag{3}$$

All subsets have size p, and M_N is the adjoint. Define

$$q = e^{-4\lambda}, \qquad s = \sqrt{q}, \qquad \gamma = q^{1/4}. \tag{4}$$

The preceding stage evaluated all words through degree four. It found $\varphi(hM^\dagger) = \gamma$ and a seven-dimensional degree-at-most-two quotient. The task here is the full joint law, not another derivation of the pure-M dressed oscillator.

The literature inputs are the exact oriented CAR sign organization and pair-intersection method of complex DSSYK [1, Sec. 3.2–3.3], and the state-adapted perspective of Rajgadia–Xu [2]. Their separate matter disorder is not substituted for our same-J block. Our coefficient proof below keeps the fixed Slater occupation instead of a trace and bounds the discarded clusters after the required normalization.

2 The limiting word rule

Write a word in $h, M^\dagger, M$ from left to right. For each Gaussian pairing π, call an hh edge G. Every other edge is selected, S, and its M endpoint fixes its creation direction. An edge with equal non-h endpoints is incompatible. A selected edge is forward when its creation acts first, from the right; otherwise it is backward. Let $b(\pi)$ count mixed hM edges. For a pairing with all selected edges forward, let

$$\nu(\pi) = \#\{\text{crossing edge pairs}\} + \#\{\text{nested edge pairs with selected inner edge}\}. \tag{5}$$

Disjoint pairs and nestings with general inner edge add zero.

Theorem 1. *For each fixed word w of length $2k$,*

$$\lim_N \varphi_N(w) = \sum_{\pi \text{ compatible, all selected edges forward}} \gamma^{b(\pi)} q^{\nu(\pi)}. \tag{6}$$

Odd moments vanish. The empty word has value one.

The remainder of the proof establishes this for arbitrary fixed k, including signs and the suppression of backward edges. It does not rely on inferring general factorization from fourth or sixth moments.

3 An exact single-site polynomial

Orient each Gaussian edge $e = (i, j)$, $i < j$, with right operator $X_e = C^\dagger_{P_e} C_{Q_e}$ and left $X^\dagger_e$. General edges allow arbitrary p-subsets P_e, Q_e. Forward selected edges require $P_e \subset E, Q_e \subset O$; backward ones require $P_e \subset O, Q_e \subset E$. Introduce $2k$ membership variables z_{P_e}, z_{Q_e}.

For a site of initial occupation $o \in \{0, 1\}$ and a membership subset S of those $2k$ labels, calculate its one-mode CAR matrix element by following the slots from right to left. At a right endpoint annihilate Q_e then create P_e; at a left endpoint annihilate P_e then create Q_e. Omit actions whose labels are absent from S. Set the result to zero if a selected-edge site restriction is violated. Multiply by

$$(-1)^{\sum_{e<f:\ e,f\ \text{cross}} t_e(S) t_f(S)}, \qquad t_e(S) = \mathbf{1}_{P_e \in S} + \mathbf{1}_{Q_e \in S}. \tag{7}$$

Call this coefficient $f_o(S) \in \{-1, 0, 1\}$, with $f_o(\varnothing) = 1$, and define

$$F_o(z) = \sum_{S \subset \{P_1, Q_1, \ldots, P_k, Q_k\}} f_o(S) \prod_{i \in S} z_i. \tag{8}$$

The raw Gaussian contraction is exactly

$$A_N(\pi) = [\prod_{i=1}^{2k} z_i^p]\ F_0(z)^a F_1(z)^m. \tag{9}$$

Why the CAR sign is exact

Split each number-conserving edge into its two thin, oriented fermion chords. Grouping events by site produces the sign $(-1)^{\sum_{u,v\ \text{cross}} (p^2 - |I_u \cap I_v|)}$. This is the exact grouping identity in the oriented CAR derivation [1, Eqs. (3.11)–(3.12)]; changing the trace to a diagonal occupation matrix element changes the local factors, not this permutation sign. The two thin chords within one edge are nested. Two crossing thick edges give four thin crossings, so their $4p^2$ contribution is even. The remaining overlap parity is precisely (7). Reversal phases of the ordered monomials cancel between the two endpoints. This argument includes sites belonging to three or more thin labels; they have not been dropped.

Choosing a membership subset at each of the a empty and m occupied sites gives (9). The coefficient enforces size p for every set. There is no independent-probe replacement and no positivity assumption on individual finite-size contractions.

4 A normalized coefficient bound for every cluster

Put $n = a = m = N/2$. Irrespective of edge type, a nonzero singleton can only be P_e at an empty site or Q_e at an occupied site. For general and forward selected edges these singletons have coefficient one. For backward selected edges both are forbidden. Use the fictitious singleton baseline $L_N^k = \binom{n}{p}^{2k}$ even in the backward case.

For each reservoir o and membership S with $|S| \geq 2$, let $n_{o,S}$ be its number of sites. Let

$$d_i = \sum_{o,S \ni i} n_{o,S}, \quad A_o = \sum_{|S|\geq 2} n_{o,S}, \quad U_o = kp - \sum_{i:\ r_i = o} d_i + A_o, \tag{10}$$

where $r_{P_e} = 0, r_{Q_e} = 1$. The correct singleton multiplicity of label i is $p - d_i$. Feasible configurations have $d_i \leq p$; if that singleton is forbidden, feasibility also requires $d_i = p$. After division by L_N^k, each configuration has signed weight

$$\frac{(n)_{\underline{U_0}}(n)_{\underline{U_1}}}{(n)_{\underline{p}}^{2k}} \prod_i (p)_{\underline{d_i}} \prod_{o,S} \frac{f_o(S)^{n_{o,S}}}{n_{o,S}!}. \tag{11}$$

This follows directly from the multinomial expansion of (9); singleton factorials have been absorbed into $(p)_{\underline{d_i}}$.

For fixed k, take $kp < n/2$. Since $|f_o(S)| \leq 1$, $(p)_{\underline{d}} \leq p^d$, and the denominator falling factorials differ from n^{2kp} by a uniformly bounded exponential, the absolute value of (11) is at most

$$C_{k,\lambda} \prod_{o,S} \frac{\alpha_{S,N}^{n_{o,S}}}{n_{o,S}!}, \qquad \alpha_{S,N} = n(p/n)^{|S|}. \tag{12}$$

Here $C_{k,\lambda}$ can be chosen uniformly along the sequence with p^2/N in a compact positive interval. In particular,

$$\sum_{o,|S|\geq 2} \alpha_{S,N} = O_k(1), \qquad \sum_{o,|S|\geq 3} \alpha_{S,N} = O_k(p^3/N^2) = o(1). \tag{13}$$

Summing (12) over all counts gives an exponential series. The total absolute weight with at least one higher cluster is $o(1)$, and the tail with more than R defects is bounded by $C \sum_{j>R} A^j/j!$ for a fixed finite A. This controls signed cancellations and rare large-overlap configurations after normalization.

Forward limit and backward suppression

For any fixed collection of double-cluster counts, the factorials in (11) have limit

$$e^{-2\lambda k(k-1)} \prod_{o,|S|=2} \frac{(2\lambda f_o(S))^{n_{o,S}}}{n_{o,S}!}. \tag{14}$$

Indeed $U_o = kp + O(1)$, and $\log(n)_{\underline{r}} = r\log n - r(r-1)/(2n) + O(r^3/n^2)$ uniformly for $r \le kp$; also $(p)_{\underline{d}}/p^d \to 1$ for fixed d. The bound on defect tails justifies summing these limits:

$$\frac{A_N(\pi)}{L_N^k} \longrightarrow \exp\left[2\lambda\left(\sum_{o,|S|=2} f_o(S) - k(k-1)\right)\right]. \tag{15}$$

For a backward edge, at least one forbidden correct singleton forces $d_i = p$. Since each defect contains label i at most once, the number of defect sites is at least p. Its total normalized weight is bounded by $C\sum_{j\ge p} A^j/j! \to 0$. This includes general H edges, for which the old pure-ladder backward estimate alone would not suffice.

The complete quadratic local table

For an all-forward graph the sum of double-cluster coefficients is obtained from the following one-site CAR cases:

Membership sources	Sum over allowed reservoir and two labels
Within a general edge: P_e, Q_e	1
Within a selected edge	0
Two disjoint edges	2
Two crossing edges	0
Nested edges, inner general	2
Nested edges, inner selected	0

For disjoint edges the surviving pairs are P_eP_f at empty sites and Q_eQ_f at occupied sites. For nested edges the two survivors instead put the inner edge's annihilator on an empty site or its creator on an occupied site; an inner selected edge forbids both. Crossings have no alternating two-label local history. Within a general edge only an initially occupied site carrying both P_e, Q_e survives.

Writing k_G for the number of general edges, the sum is $k_G + k(k-1) - 2\nu(\pi)$. Equation (15) is therefore $e^{2\lambda k_G} q^{\nu(\pi)}$. A word with n_h Hamiltonian endpoints has normalization $L_N^k (D_N/L_N)^{n_h/2}$, with $D_N/L_N \to e^{2\lambda}$ and $n_h = 2k_G + b(\pi)$. The remaining factor is $e^{-\lambda b} = \gamma^b$. Summing the finitely many pairings proves Theorem 1.

5 Uniform finite-size bounds before taking time limits

Fixed-degree convergence by itself does not justify a real-time series. Here the actual finite operators have a stronger bound. Expand all of them in the same independent standard real Gaussian coordinates x_α:

$$h_N = \sum_\alpha x_\alpha A_\alpha, \qquad M_N = \sum_\alpha x_\alpha B_\alpha. \tag{16}$$

On the m-particle sector,

$$\sum A_\alpha^\dagger A_\alpha = \sum A_\alpha A_\alpha^\dagger = I, \qquad \sum B_\alpha^\dagger B_\alpha \le I, \quad \sum B_\alpha B_\alpha^\dagger \le I. \tag{17}$$

The first identity is the exact sector variance D_N. For the selected block the two sums are diagonal on occupation states. On a state with x particles in E their eigenvalues, in either order, are

$$\frac{\binom{x}{p}^2}{L_N}, \qquad \frac{\binom{m-x}{p}^2}{L_N}, \qquad 0 \le x \le m, \tag{18}$$

both at most one. The circular Gaussian real and imaginary parts have their $1/\sqrt{2}$ factors; hence no extra factor two appears. Adjoints obey the same row/column bounds.

Every Wick contraction is a contraction

For a coefficient family C_α, opening a Gaussian edge uses $v \mapsto \sum_\alpha e_\alpha \otimes C_\alpha v$; its norm is $\|\sum C_\alpha^\dagger C_\alpha\|^{1/2}$. Closing it uses $\sum e_\alpha \otimes v_\alpha \mapsto \sum C_\alpha v_\alpha$, with row norm $\|\sum C_\alpha C_\alpha^\dagger\|^{1/2}$. Keep tensor slots for all open Gaussian indices and permute them unitarily to close the required edge. Following any pairing from right to left reproduces its matrix product in its original order. By (17) its operator norm is at most one. Thus for every finite N and every word of length $2r$,

$$|\varphi_N(w)| \le (2r-1)!!. \tag{19}$$

This proof allows crossings and the shared disorder. It is not a scalar-independence approximation. For a word v of length d,

$$\|v\Omega_O\|_{L^2(J)} \le F_d, \qquad F_d = \sqrt{(2d-1)!!}, \quad F_0 = 1. \tag{20}$$

The bound is uniform in both N and degree. Unlike the cluster proof, it has no degree-squared exponential prefactor.

6 Joint representation and finite-time convergence

Define φ on the free star-polynomial algebra by Theorem 1. It is positive because for every fixed polynomial P,

$$\varphi(P^*P) = \lim_N \varphi_N(P_N^* P_N) \geq 0. \tag{21}$$

Quotient by its null left ideal and complete to obtain the cyclic GNS space. Left multiplication defines $h = h^*$ and the mutually adjoint-on-the-core $M, M^\dagger$. This provides an all-degree constructive representation through its Gram matrices and multiplication, without assuming a finite closed Hamiltonian.

If P has degree at most d and coefficient one-norm C_P, (20) gives

$$\|h^n P\Omega\| \leq C_P F_{n+d}, \qquad \sum_{n\geq 0} \frac{|z|^n}{n!} \|h^n P\Omega\| < \infty \quad (z \in \mathbb{C}). \tag{22}$$

Every core vector is entire analytic for h. Nelson's analytic vector criterion makes this symmetric Hamiltonian essentially self-adjoint; we use the standard theorem as recalled in [3, Theorem 4.8]. Its closure generates the unitary e^{ith}. Polynomial insertions on evolved core vectors are defined by the same graph-norm-convergent series; the moment bound remains valid with their additional fixed degree.

More concretely, for fixed star-polynomials $P_0, \ldots, P_\ell$, fixed complex $z_1, \ldots, z_\ell$, and $d = \sum_j \deg P_j$, the ordered correlator

$$\varphi_N\left(P_{0,N} e^{z_1 h_N} P_{1,N} \cdots e^{z_\ell h_N} P_{\ell,N}\right) \tag{23}$$

converges to the series obtained by replacing every polynomial word coefficient with (6). With $T = \sum_j |z_j|$ and $C = \prod_j C_{P_j}$, the absolute tail after total exponential degree $R-1$ is bounded uniformly in N by

$$C \sum_{n\geq R} \frac{T^n F_{n+d}}{n!}. \tag{24}$$

The multinomial theorem combines all time powers into $T^n/n!$. Their successive majorant ratio is

$$\rho_n = \frac{T\sqrt{2(n+d)+1}}{n+1} \longrightarrow 0. \tag{25}$$

It decreases with $n \geq 0$. When $\rho_R < 1$, (24) is at most $CT^R F_{R+d}/[R!(1-\rho_R)]$. First take $N \to \infty$ at fixed R, then $R \to \infty$; the bound is uniform on compact complex-time sets. Physical string time uses $H_N = \sqrt{w_N D_N} h_N$, where $w_N D_N \to J_s^2/\lambda$. No time growing with N is claimed.

7 Six-point consequences and independent finite checks

The familiar Hamiltonian moment is recovered:

$$\varphi(h^6) = 5 + 6q + 3q^2 + q^3, \qquad \varphi(M^3(M^\dagger)^3) = 6q^3. \tag{26}$$

The actual disorder complement remains $g = h - \gamma(M + M^\dagger)$, with $\varphi(g^2) = 1 - s$. Its sixth moment from the joint law is

$$\frac{\varphi(g^6)}{(1-s)^3} = 5 + 6s + 14q + 10sq + 13q^2 + 7sq^2 + 6q^3. \tag{27}$$

Together with the preceding fourth moment $2 + s + 2q$, this quantifies why a standard independent q mode is not the Hamiltonian complement.

The new integer CAR driver checks all fifteen three-edge pairings for $(N, p) = (4, 1), (6, 1), (4, 2)$. There are 1,461,120 independent Gaussian label assignments. Each raw matrix element is compared with the single-site product including (7). The first two systems contain respectively 8 and 72 negative nonzero contributions; all are reproduced. Independently extracting (9) verifies all 3^3 general/forward/backward edge specifications for each pairing and fixture: 1,215 coefficients. The quadratic local table is checked for all pairings and forward edge types through four edges, 1,814 cases.

A separate exact symbolic driver lists all 729 sixth-word polynomials and verifies (27). Rational Gram elimination at $\gamma = 1/2, 2/3$ gives rank seven through degree two and rank fifteen through degree three, with positive nonzero pivots. The latter rank is a two-parameter-point check, not a proof of generic rank fifteen or a universal binary-tree representation. Finite row/column covariance counts and the scalar time-tail majorant are separately recorded. These checks support the specific implementations; the arbitrary-degree and time claims rest on the proofs above.

The source, parameters, versions, input hashes and three scientific outputs are stored in `results/susskind-slater-joint-words/`. The evidence manifest also binds the final LaTeX, PDF, two-pass build logs and every rendered page. Old successful scientific outputs were not regenerated or overwritten.

8 What this makes possible next

The same-J joint dynamics is now specified by a positive all-word law and a convergent finite-time construction in the annealed Slater state. It can be used to evaluate dynamical commutators, spectral measures with selected insertions and state-preparation kernels. It also supplies a precise object against which proposed canonical or bulk representations can be tested, beyond matching two or four moments.

The next constructive task is to turn the Gram/multiplication description into a transparent transfer representation and to compute physically useful kernels efficiently. The degree-three rank check suggests an economical organization, but no such organization is assumed in this paper. The four second-layer directions of the preceding stage must all survive any proposed representation with their known Hamiltonian coefficients.

The microscopic Slater vector is not stationary under h. A physical observer state, its preparation and its relation to the original static-patch observables must therefore be specified before identifying particle masses or spatial generators. The joint moment limit here is annealed; concentration for the full joint algebra is not asserted. It has fixed positive λ and fixed time windows, and says nothing about recurrences at times growing with N. Nor is this a derivation of a finite-G gravitational scattering amplitude. These remaining questions preserve the original de Sitter research target.

References

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